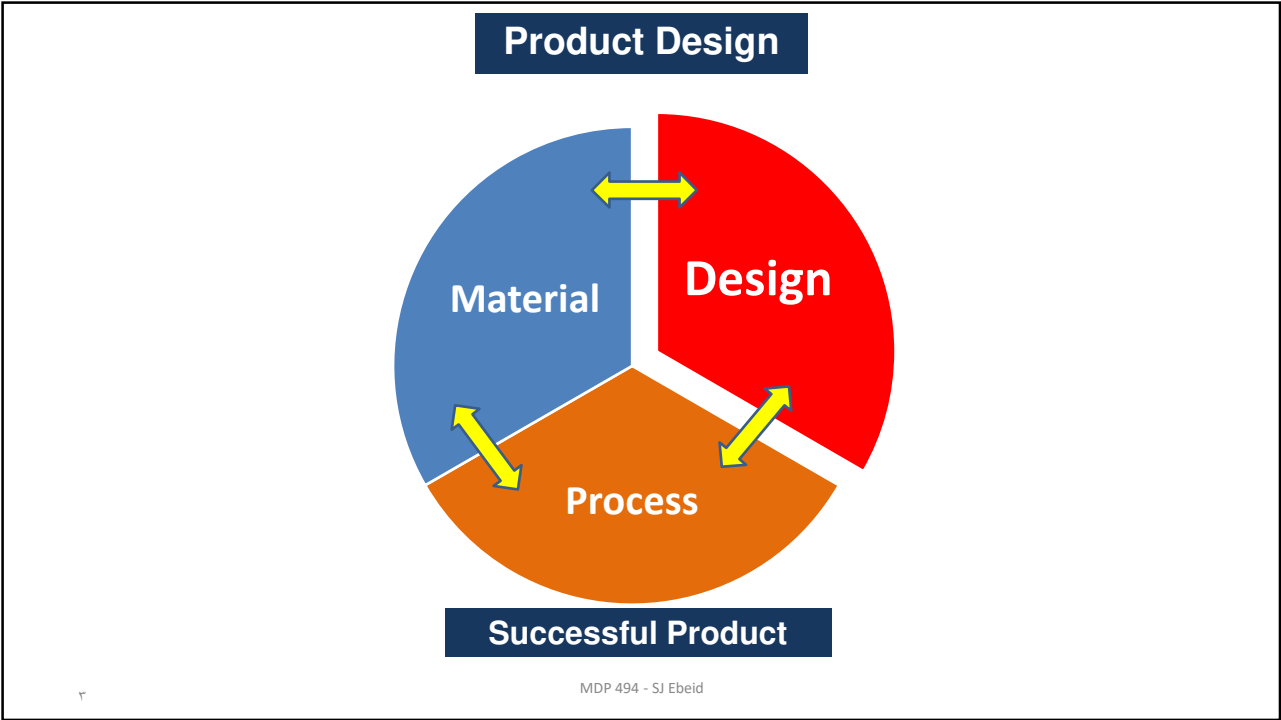




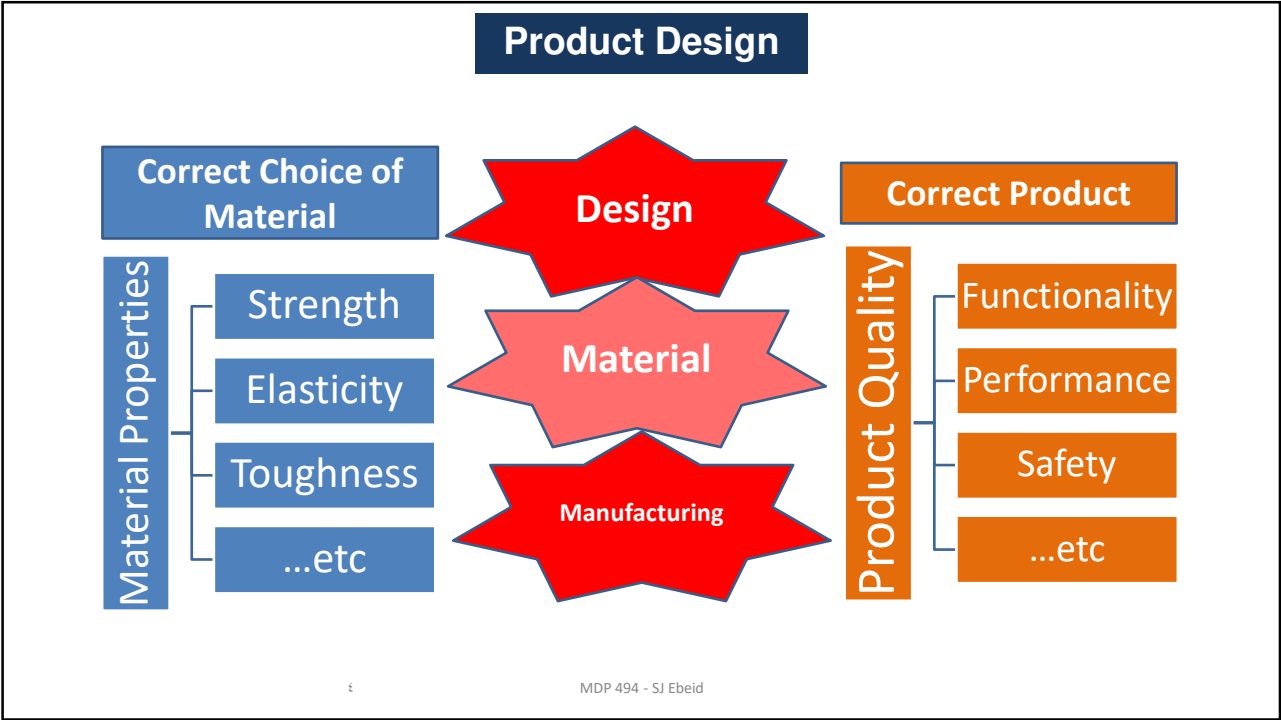
1



2



3



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Products



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5

Products



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6

Products



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8



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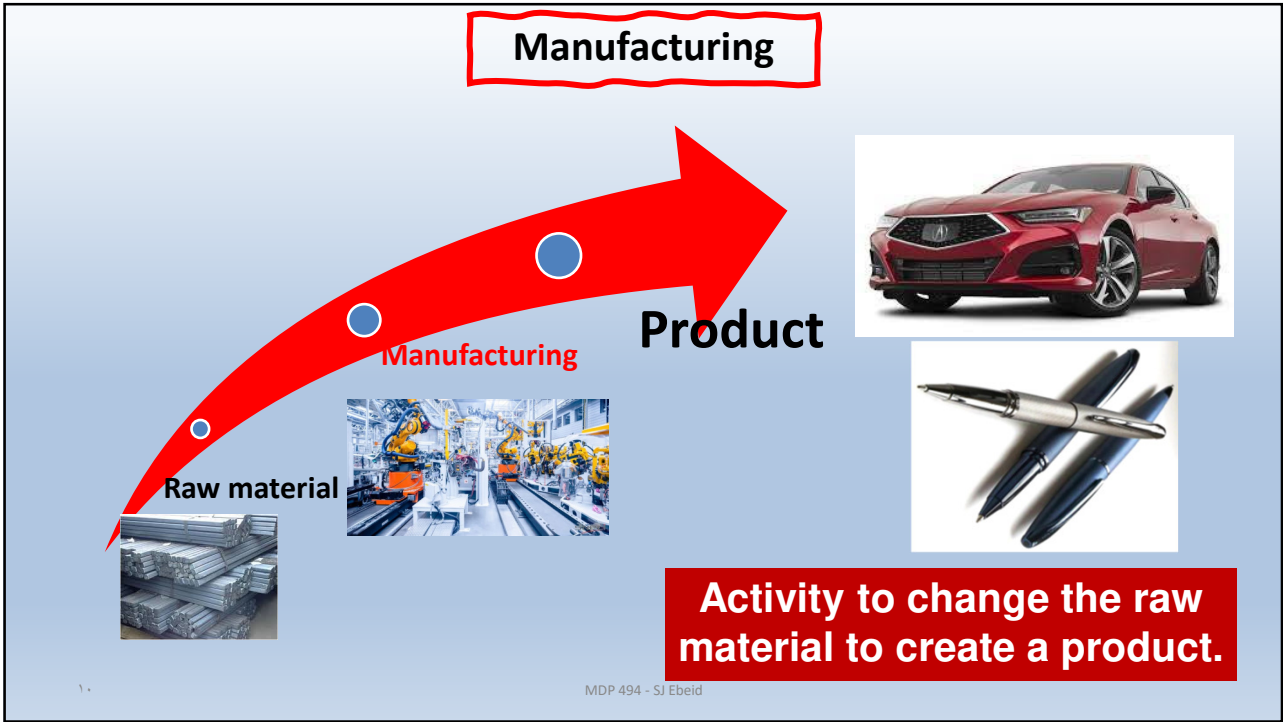
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Manufacturing



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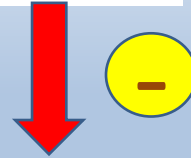
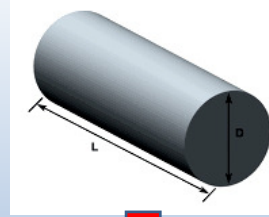


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Machining

Machining is the removal of material to obtain a product.



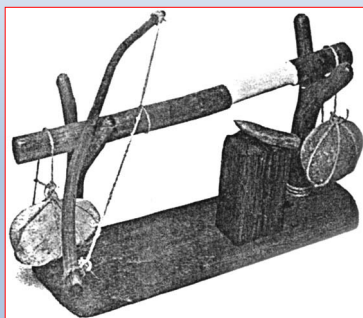
WP:
Shape
Size
Surface

11

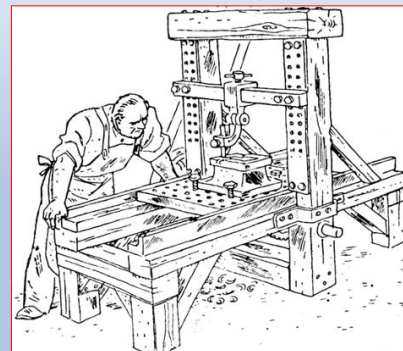
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Old Machining Techniques



A very old Lathe Machine



A very old Planing Machine

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12

Old Machining Techniques



Lathe Machine - 1920

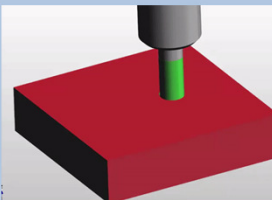
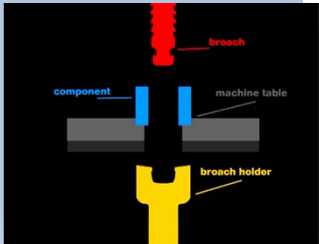
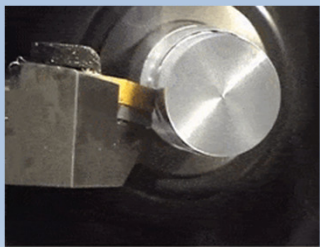
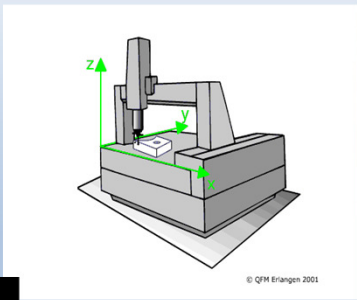


Drilling Machine

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Machining Processes



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Exceptional Products

Shapes & Sizes

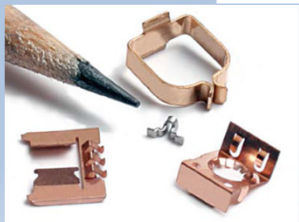
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Exceptional Products

Shapes - Sizes

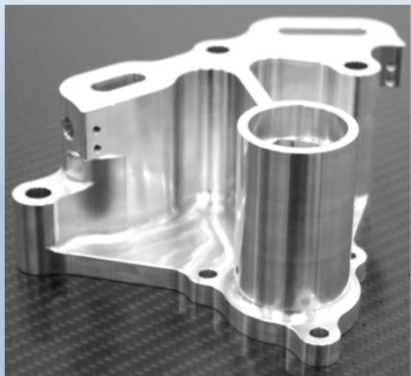


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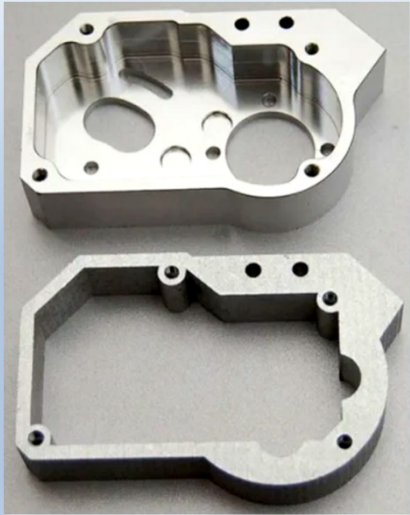
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Very Hard & Complicated Components



1Y



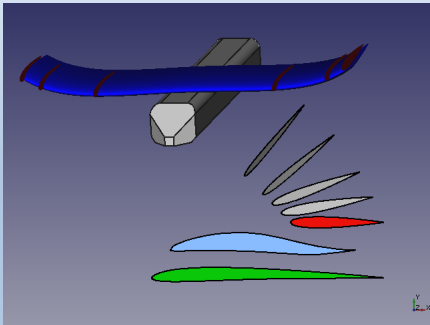
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Very Hard & Complicated Components



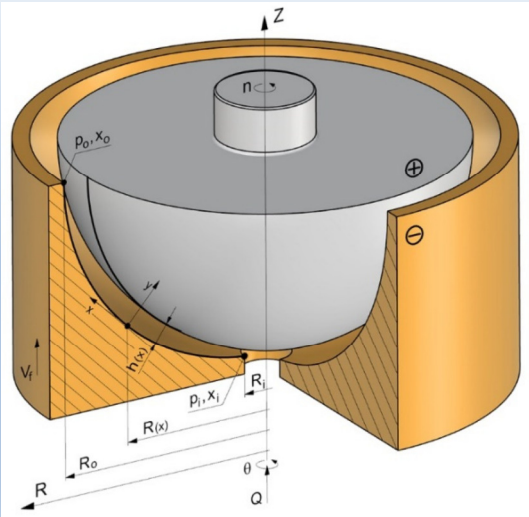
1A



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Very Hard & Complicated Components



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19

Very Hard & Complicated Components



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Non-Conventional Machining

New Forms of Energy

Mechanical Energy

Light Energy

Chemical Energy

Thermal Energy

Electrical Energy



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Concurrent Design

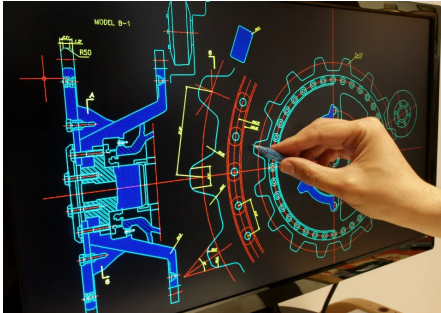
Traditional



We design it

You build it

CAD



We & you design it and build it

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Prof. Samy J. Ebeid

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Parametric Design

* Before the use of “*parametrics*”, editing the shape was not an easy task for designers. To modify a 3D solid, the designer had to change the length, the breadth and the height.

* After the use of **parametric modelling**, the designer needs only to alter **one parameter**; the other two parameters get adjusted automatically.

Parametric means that the model is guided by parameters, which can include dimensions, wall thicknesses, hole diameters and depths, and so on.

Parametric

is a term used to describe a dimension's ability to change the shape of model geometry if the dimension value is modified.

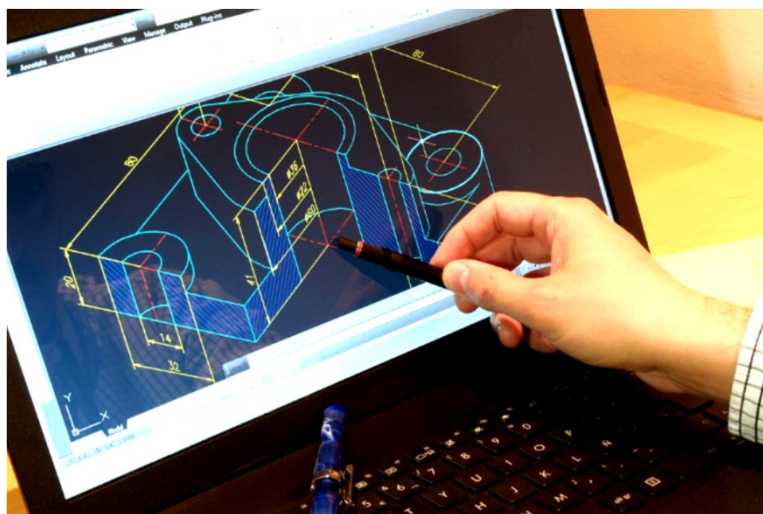


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CAD



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CAD

CAD involves any type of design activity that makes use of a computer to:

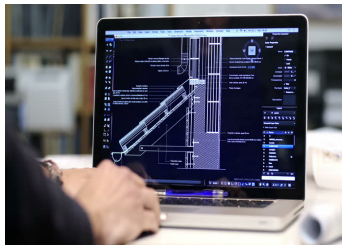
- create
- develop
- analyze
- optimize
- modify an engineering design.

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Effect of CAD on the Design Process



Faster design process ... minimizes trial & error

Increases productivity ... efficient manufacturing

Improves communication between designers

Easier modification

Includes libraries for standard parts

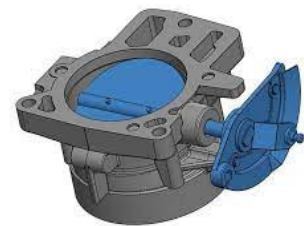
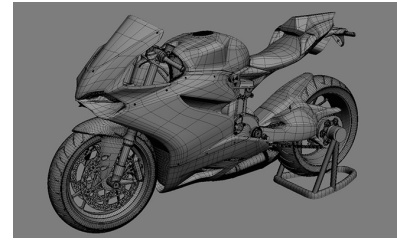
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Advantages of Modeling by CAD

- 3D models are easier to interpret.
- Simulation under real-life conditions.
- Less expensive than building a physical model.
- 3D models can be used to perform FEA (stress, deflection, thermal.....).
- 3D models can be used directly in manufacturing.
- Can be used for presentations and marketing.



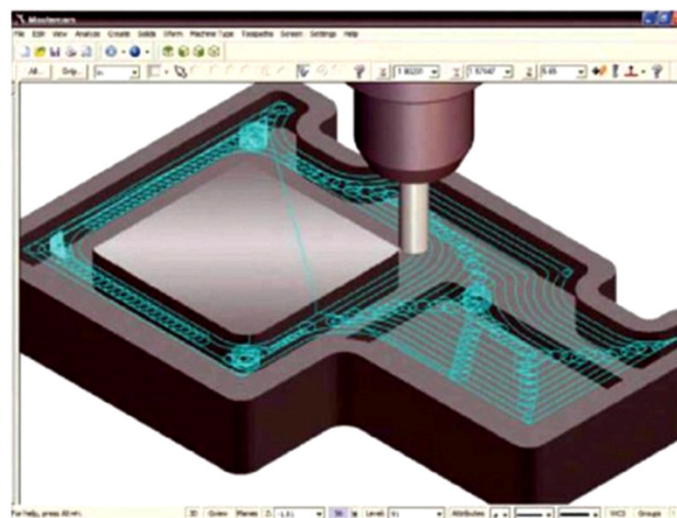
Modeling by CAD, is followed by analysis using CAE for design validation.

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CAM



Y/A

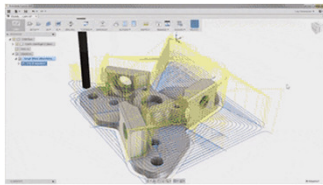
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CAM

Computer Aided Manufacturing is the use of software and computer-controlled machinery to **automate a manufacturing process**.

CAM uses software to **translate** drawings and data into detailed instructions that can drive the cutting tools.



The data from **CAD** drawings can be extracted and used to create the **G-Code** used by **CNC** in machine tools for manufacturing components.

G-code is a computer numerical control programming language, used in **CAM** to control automated machine tools.

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